

Word Embeddings and Semantic Geometry

DATA 443/643 · Week 1, note 1.4

After Stanford CS224N Lecture 2: distributional meaning, skip-gram softmax, then evaluation.

Lecture plan

- 1 One-hot is a bad geometry
- 2 How a corpus becomes a distributional vector
- 3 Skip-gram softmax, then what you keep
- 4 Analogies, evaluation, and bias as geometry

Key goal: a word can be a dense vector learned from its neighbors; you can write $P(o \text{ given } c)$; projects report extrinsic numbers.

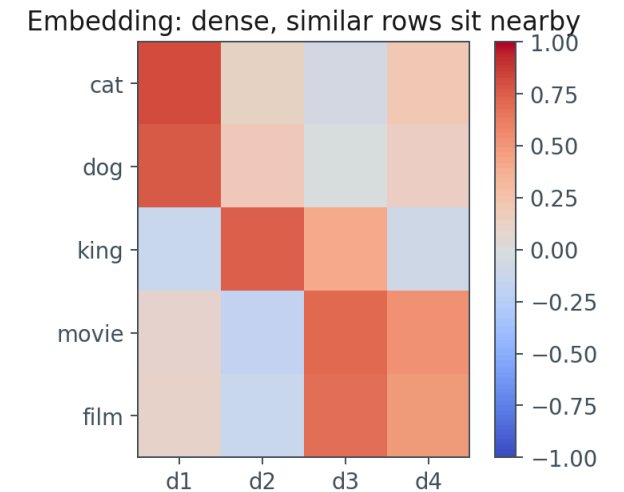
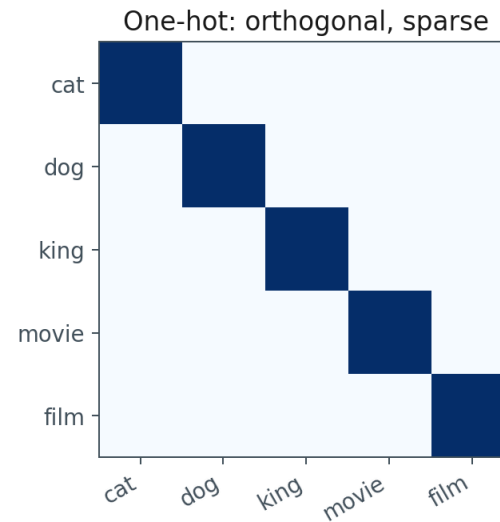
First time: what / why / formula

- What / why. A short dense vector per word (or subword). Similar usage $\rightarrow$ nearby vectors. One-hot geometry is orthogonal: *film* and *movie* have inner product 0. You need a geometry you can learn.
- Architecture. A table $E \in \mathbb{R}^{V \times d}$. Row i is the vector for token i . Skip-gram trains it with a softmax classifier you then throw away.
- Formula. $P(w_o \mid w_c) = \exp(u_o^\top v_c) / \sum_w \exp(u_w^\top v_c)$. Cosine: $u^\top v / (\|u\| \|v\|)$.
- Tradeoffs. + Analogies and nearest neighbors as a debug tool. – Static vectors (one vector per word type); bias in the corpus becomes geometry; Week 3 will replace this with contextual states.

How. Windows of neighbors. Skip-gram: from the center, predict a neighbor. Keep the center rows.

Problem with words as discrete symbols

- film and movie are orthogonal.
- No natural similarity for one-hot vectors.
- A new word has no coordinate.



Three words that are not synonyms

Distributional

Similar contexts, similar meaning.
Harris; Firth; Jurafsky Ch. 5.

Distributed

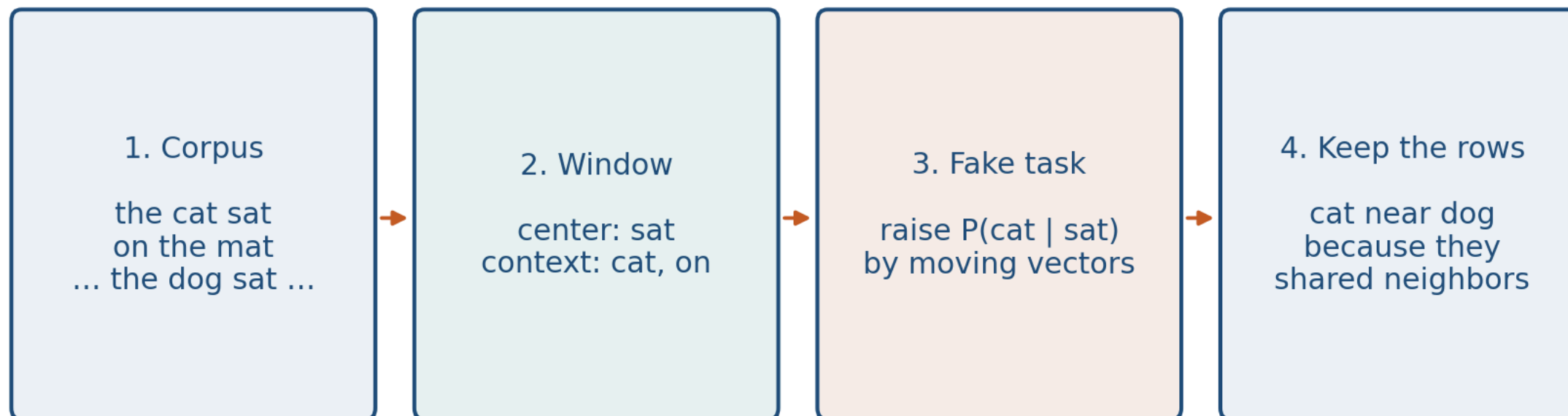
The code is a dense vector, not a one-hot.

Embedding

The lookup: `nn.Embedding`, a row of a V by d table.

How a distributional embedding is obtained

Distributional embeddings are obtained from running text, not from a thesaurus



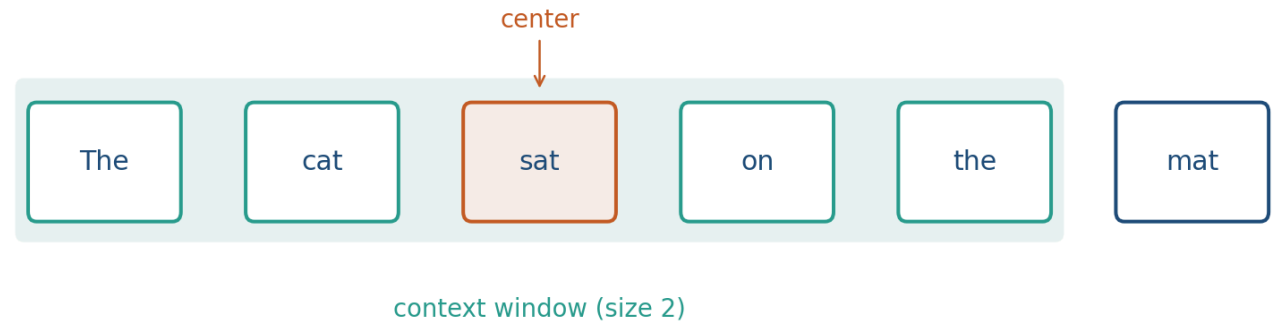
Corpus, then a window, then a fake prediction task. Words that shared neighbors sit nearby. Those rows are the embeddings.

Firth: you shall know a word by the company it keeps. Word2Vec is one algorithm for that. A thesaurus is not.

Skip-gram: from the center, predict a neighbor

- Fake task on a window.
- CBOW: from neighbors, predict the center.
- Logits $z_w = u_w^\top v_c$.

Skip-gram: predict each neighbor from the center word



Word2Vec prediction function

$$P(w_o \mid w_c) = \frac{\exp(u_o^\top v_c)}{\sum_w \exp(u_w^\top v_c)}$$

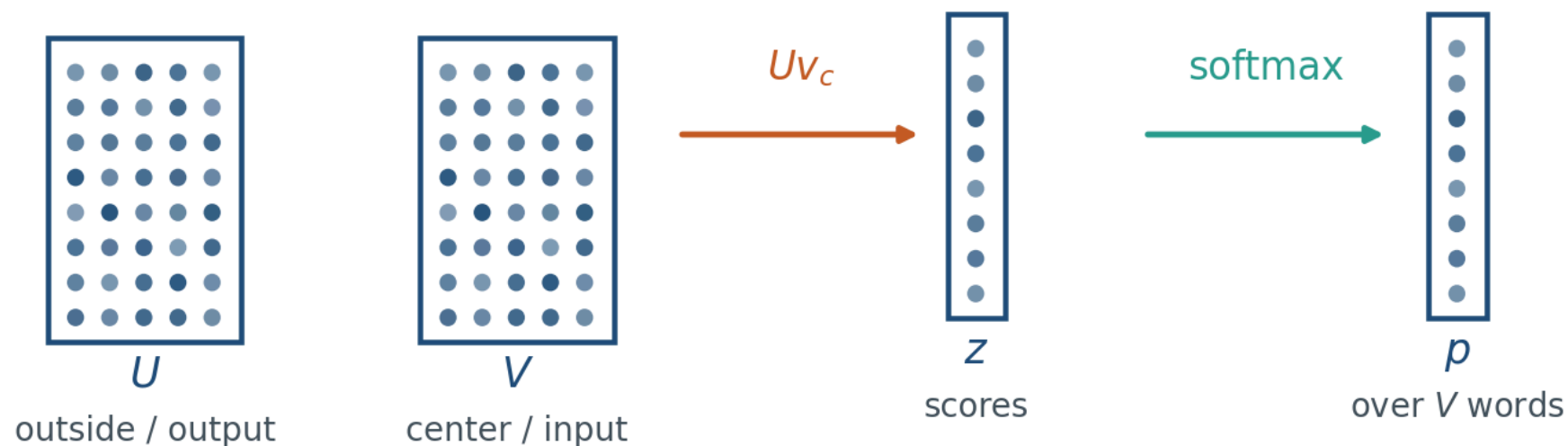
1. Dot product scores similarity of o and c.

2. Exp makes every score positive.

3. Normalize over the vocabulary. That is softmax.

Skip-gram is a table, a product, then a softmax

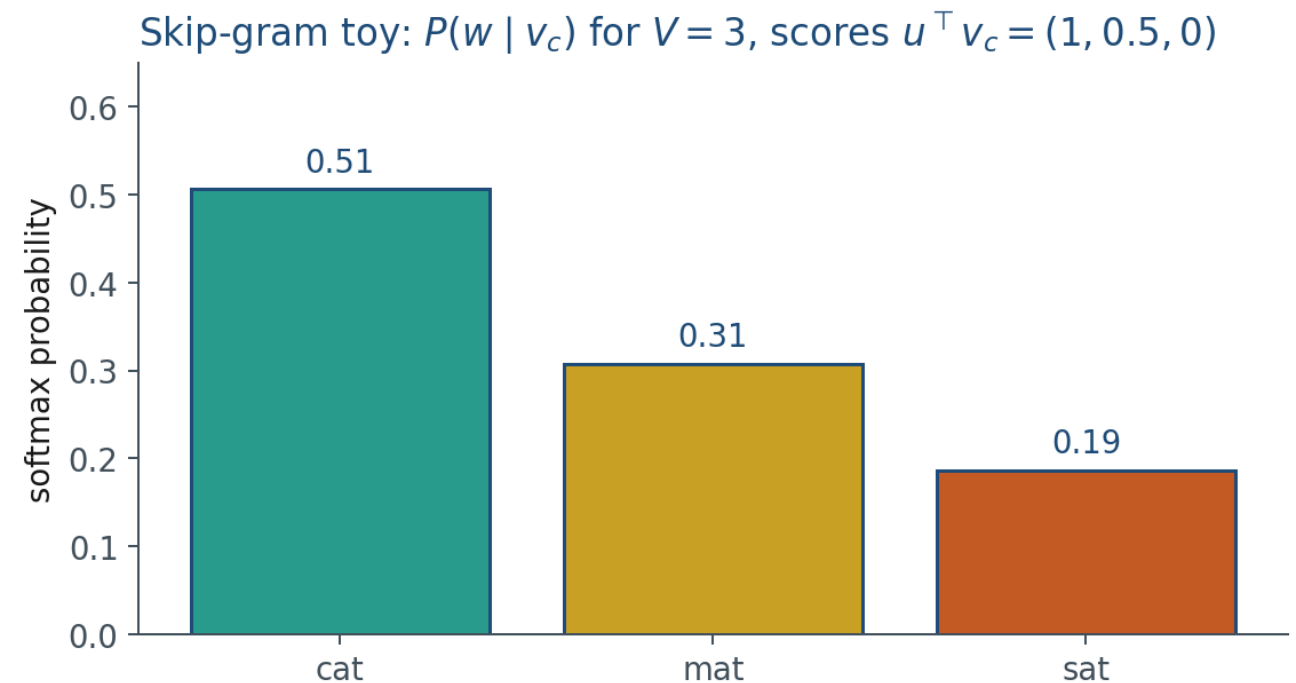
Skip-gram is a lookup, a dot product, then a softmax — not a mysterious net



Keep V (center rows). Throw away the softmax. A transformer keeps transforming those vectors.

A three-word softmax you can finish by hand

- Scores $(1, 0.5, 0)$ for cat, mat, sat.
- $p \approx (0.51, 0.31, 0.19)$.
- Mikolov 2013a §§1–3. Negative sampling is cited, not derived.



Counting vs. predicting

Count

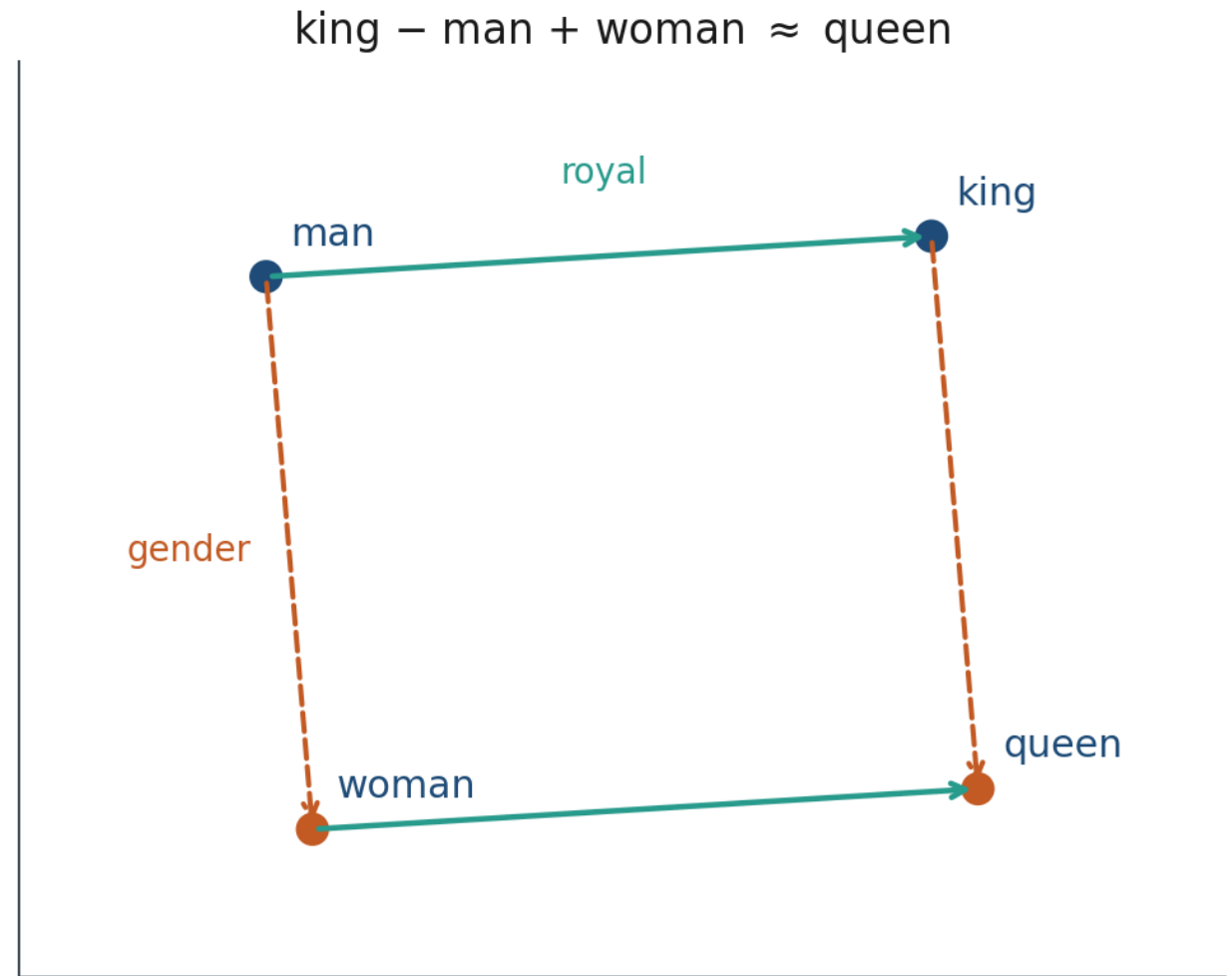
Build a co-occurrence matrix, then factor it (LSA / GloVe). Classical vector semantics can start here.

Predict

Skip-gram predicts context. Same distributional idea; same training loop as an LLM. No SVD homework this week.

Direction can mean something

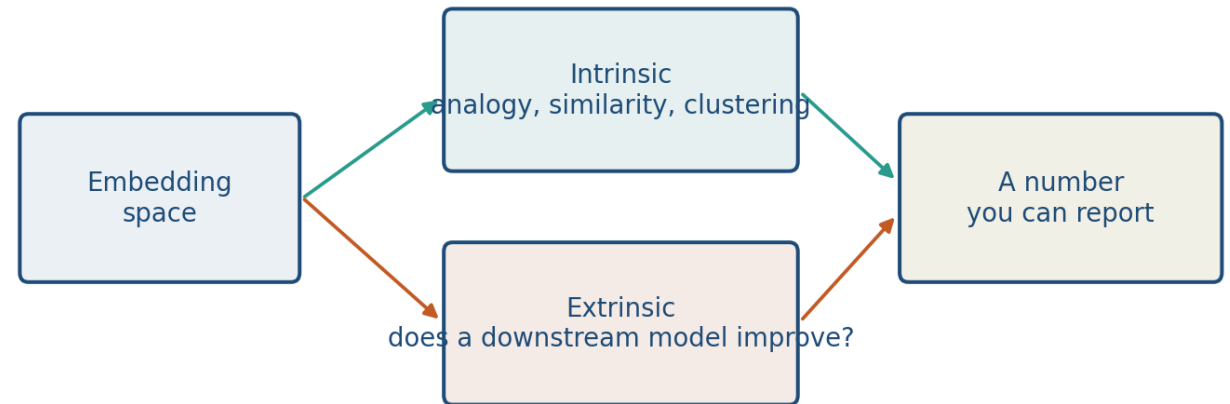
- $\text{king} - \text{man} + \text{woman} \approx \text{queen}$.
- Cosine: $u^T v / (\|u\| \|v\|)$. Length tracks frequency.
- Nearest neighbors debug. They are not a project metric.



How to evaluate word vectors

- Intrinsic: analogies, similarity, clustering.
- Extrinsic: does a downstream model improve?
- They need not agree. Projects report extrinsic.

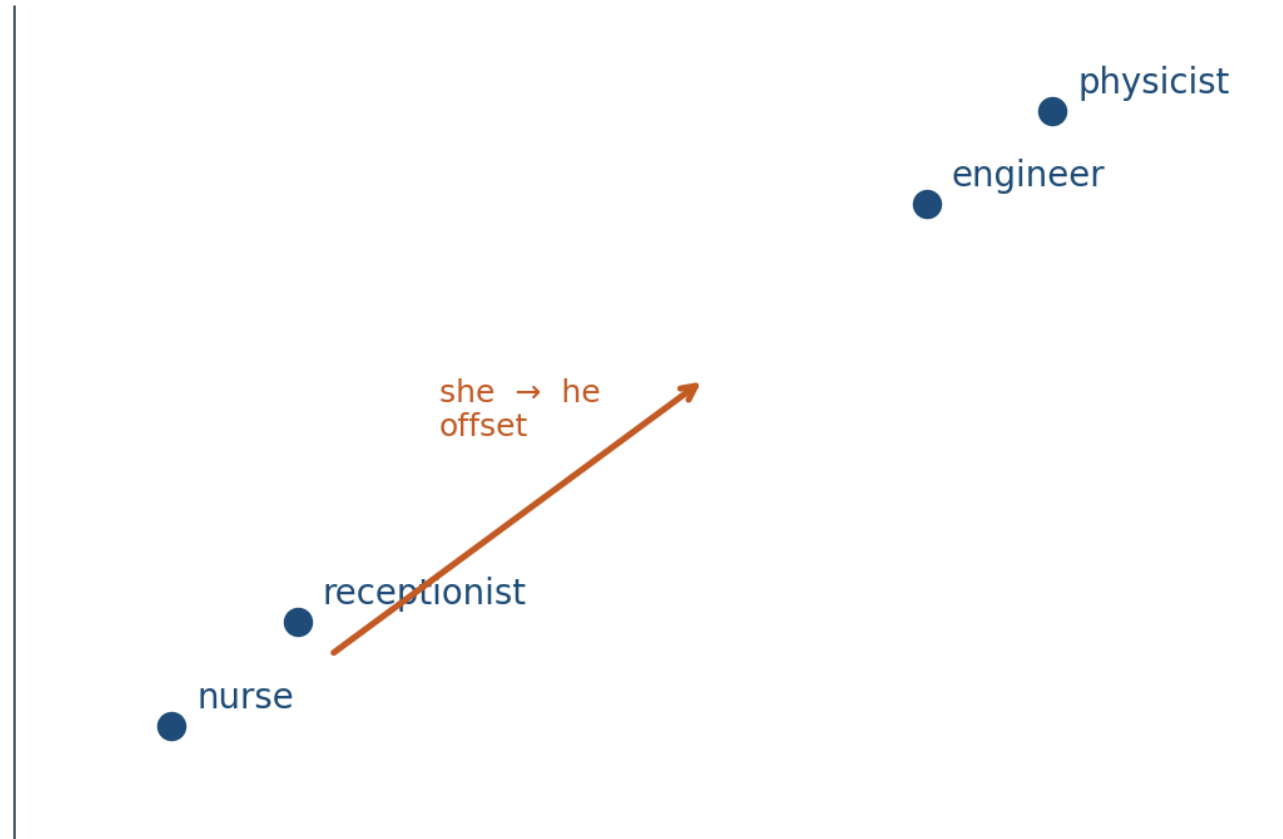
Two ways to score embeddings. They need not agree.



Bias is also geometry

- Offset $o = \vec{he} - \vec{she}$.
- Score $v^T o / \|o\|$. Larger sits closer to he.
- Lab 1: constructed 2-D first, then the same probe on a real model later.

Geometry can encode stereotypes as well as meaning



What to take from this lecture

Write it

Skip-gram softmax from the center vector. Keep those rows; throw the classifier.

Report it

Intrinsic numbers debug the space.
Extrinsic numbers go in the project.

Probe it

The same geometry can encode a stereotype. Lab 1 measures that on purpose.